

IoT-Based Smart Gas Leakage Detection and Emergency Alert System

*Project Report Submitted in partial fulfilment of the requirements
for the degree of Bachelor of Technology from Maulana Abul
Kalam Azad University of Technology, West Bengal (formerly
known as West Bengal University of Technology)*

Prepared by:

- Rudraksha Karfa (34900322020)
- Pritam Adhikary (34900322053)
- Malay Patra (34900322044)
- Buddhadeb Sarkar (34900322054)
- Basudeb Karmakar (34900322046)
- Avijit Das (34900323068)

Under the Guidance of

Dr. Palash Das

**Department of Electronics and Communication Engineering
Cooch Behar Government Engineering College**

Cooch Behar, West Bengal

2022-2026

Certificate of Recommendation

It is hereby recommended to consider the project report entitled "**IoT Based LPG Gas Leakage Detection and Emergency Alert System.**" submitted by **Rudraksha Karfa** (Roll no: 34900322020), **Pritam Adhikary** (Roll no: 34900322053), **Malay Patra** (Roll no: 34900322044), **Buddhadeb Sarkar** (Roll no: 34900322054), **Basudeb Karmakar** (Roll no: 34900322046), **Avijit Das** (Roll no: 34900323068), for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering from Cooch Behar Govt. Engineering College affiliated to Maulana Abul Kalam Azad University of Technology (formerly known as West Bengal University of Technology).

Dr. Palash Das
Project Guide

Dr. Sourav Chakraborty
HOD, ECE

Certificate of Approval

It is hereby approved that the project report entitled "**IoT Based LPG Gas Leakage Detection and Emergency Alert System.**" submitted by **Rudraksha Karfa** (Roll no: 34900322020), **Pritam Adhikary** (Roll no: 34900322053), **Malay Patra** (Roll no: 34900322044), **Buddhadeb Sarkar** (Roll no: 34900322054), **Basudeb Karmakar** (Roll no: 34900322046), **Avijit Das** (Roll no: 34900323068), for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering from Cooch Behar Govt. Engineering College affiliated to Maulana Abul Kalam Azad University of Technology (formerly known as West Bengal University of Technology).

Board of Examiners

Name

Signature

.....

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Project Members

Roll No.	Name	Signature
34900322020	Rudraksha Karfa	
34900322053	Pritam Adhikary	
34900322044	Malay Patra	
34900322054	Buddhadeb Sarkar	
34900322046	Basudeb Karmakar	
34900323068	Avijit Das	

CONTENTS

Title	Page
Acknowledgement	I
Abstract	II
Contents	III
List of Figures	IV
Chapter One	1
Introduction	1
Preface	1
Critical Systems	2
Safety Critical	2
Business Critical	2
Mission Critical	2
Security Critical	3
Problem Statement	3
Proposed Solution	3
The Objectives	3
Methodology	4
Thesis layout:	4

Chapter Two	5
background and related work	5
2.1 Background	5
2.2 Safety Critical System	6
2.2.1 Process Safety System or Process Shutdown System (PSD)	6
2.2.2 Safety Shutdown System	7
2.3 Detectors	8
2.3.1 Gas Detectors	9
2.3.2 Gas Sensors	11
2.4 Alarm Unit	13
2.5 System Component Modules Internet of Things (IoT)	24
2.5.1 WiFi Module (ESP32 / NodeMCU)	18
2.5.2 3.7V Battery & Accessories	19
2.5.3 Passive Electrical Elements & Indication	21
2.5.4 Buzzer	22
2.6 Internet of Things	24
2.6.1 The Cloud	25
2.7 Related Work	28
Chapter Three	33
Methodology	33
3.1 Overview	33
3.2 Simulation	34
3.3 System Interfacing	35

3.3.1 Interfacing Gas Sensor (MQ-6) with Arduino	35
3.3.2 Interfacing Flame and Temperature Sensors with Arduino	35
3.3.3 Interfacing the Buzzer with Arduino	37
3.3.4 Interfacing the Light Emitting Diode (LED) (Red, Green) with Arduino	37
3.3.5 Interfacing the Wireless Module (GSM900) with Arduino	37
3.4 System Operation	38
3.5 System Work Flow	40
Chapter Four	41
System result	41
4.1 System Operating Condition	41
4.1.1 Fire is Detected	41
4.1.2 Gas is Detected	42
4.1.3 Fire and Gas is Detected	42
4.1.4 No Detection	43
Chapter Five	48
Conclusion and Recommendation	48
5.1 Conclusion	48
5.2 Recommendation	48
References	49
Appendix	50

LIST OF FIGURES

Title	Page
Figure 2-2: Gas Sensor (MQ-6)	11
Figure 2-5: Alarm Unit	13
Figure 2-3: 4. 3.7 BATTERY	16
Figure 2-12: WiFi Module (ESP32)	18
Figure 2-13: 3.7V Battery	19
Figure 2-14: Battery Case	20
Figure 2-15: Step-Up Boost Converter	20
Figure 2-16: Charging Module	20
Figure 2-17: Jumper Wires	21
Figure 2-18: 3D Printed Box	21
Figure 2-19: 220 Ohm Resistors (x2)	21
Figure 2-20: Switch	22
Figure 2-21: Red LED and Green LED	22
Figure 2-18: Buzzer	22
Figure 2-21: Internet of Things (IoT) System	25

Figure 2-22: The Cloud	25
Figure 2-23: Models of Cloud	27
Figure 2-24: Things-board Cloud	27
Figure 3-1: System Block Diagram	34
Figure 3-2: Simulation System	35
Figure 3-3: System Work Flow	40
Figure 4-1: Fire Detection	41
Figure 4-2: Gas Detection	42
Figure 4-3: Fire and Gas Detection	43
Figure 4-4: No Detection	44
Figure 4-5: SMS Gas Alarm	45
Figure 4-6: Liquid Crystal Display (LCD) Gas Alarm	45
Figure 4-7: GSM Call Alarm	46
Figure 4-8: Liquid Crystal Display (LCD) Fire Alarm	46
Figure 4-9: Fire and Gas Alarm	46
Figure 4-10: Cool Temperature	47

ABSTRACT

The gas and fire detection system plays an important role in saving lives and properties and maintaining and monitoring a safe environment and situation. This project discusses the effective method to use the microcontroller (ESP32) to control the other components to provide a gas and fire safety system. From the project done, the system is giving quick response to current situation. System can detect gas and flame sensed by the detectors. When the sensors from each level triggered individually, the gas valve turn off, the ventilation fan, the alarm system operates and it shows in the control panel liquid crystal display alert, a message will be send to the person in charge of the factory in case of gas leakage, and make a call in case of fire detection, finally the factory power is shut down.

CHAPTER ONE

INTRODUCTION

Preface

Gas leakage leads to various accidents resulting in both material loss and human injuries. The risk of explosion, firing and suffocation are based on their physical properties such toxicity, flammability, etc. The number of deaths due to explosion of gas cylinders and gas pipelines has been increasing in recent years. The reason for such explosion is due to substandard cylinders, old valves, worn out regulators and lack of awareness in handling gas cylinders or leakage in gas pipelines due to creaks or the effect of heat. Flammable gases are used as fuel in many applications like homes, hostels, industries, automobiles, vehicles. The serious problem of this gas's leakage is that they are heavier than air; these gases do not disperse easily. It may lead to suffocation when inhaled and may lead to explosion. Due to the explosion of gases, the number of deaths has been increased in recent years. To avoid this problem there is a need for a system to detect the gas leakage. The situation should be taken before any fire/blast could happen because of its highly sensitive and flammable nature it can cause great disaster if any fire source is exposed to the gas so special countermeasure should be taken for the gas leakage. Gas leak detection is the process of identifying potentially hazardous gas leaks by means of various sensors. This paper presents a gas leakage detection and alert system to avoid fire accidents and to provide safety. The infamous Bhopal gas tragedy of 1984, which claimed the lives of thousands, is one of the major accidents due to gas

leakage[1]. And another instance, the HPCL refinery disaster destroyed the lives of many families.

Critical Systems

A critical system is a system which must be highly reliable and retain this reliability as they evolve without incurring prohibitive costs, and whose failure could threaten human life, the system's environment or the existence of the organization which operates the system.

Safety Critical

Is a system whose failure or malfunction may result in death or serious injury to people, loss or severe damage to equipment/property, environmental harm, a safety-critical system can be a control system for a chemical manufacturing plant or aircraft.

Business Critical

A system whose failure may result in the failure of some goal-directed activity and is essential to the survival of a business or organization, an example of a mission-critical system is a navigational system for a spacecraft.

Mission Critical

A mission critical system also known as mission essential equipment and mission critical application is any system that is essential to the survival of an organization and the operation of the organization is affected if the system fails or is interrupted in any way, an examples of mission critical systems include control systems for aircraft, electricity grid systems, and emergency communications systems.

Security Critical

Safety critical systems (SCS) are systems designed with the intent of curbing the effects of an accident from a hazardous event. This can be implemented in the aviation industry, the medical profession.

Problem Statement

The gas and fire safety system in Saria batteries factory which uses different toxic gases that can result in a hazardous work environment if a gas leakage due to creaks in pipelines or old valves in gas cylinders occurred, it can also lead to explosions when exposed to heat or sparks, which could result in great losses both material and human lives.

Proposed Solution

A gas detection and monitoring system based on IoT using MQ-2 and MQ-7 sensors and Arduino microcontroller, allowing remote monitoring of the system, and notifying the person in charge when the gas exceeds a threshold level or a fire is detected through SMS or a phone call.

The Objectives

- 1- activate the safety system to reduce the leakage gas and fire accident by implementing gas concentration detection and fire detection
- 2- make the work environment healthier by reducing the toxic gas

Methodology

To build system that is able to detect gas leakage, Fire detection and alerting its owner to take precautions. We start building the hardware part which consist of Flame Sensor, Gas sensors (MQ-2 and MQ-7) and temperature sensor, All these sensors are collecting the environment data as Input, the feedback from this data is shown in the actuators which are gas valve, electrical panel, fire fighting system, ventilation fan, alarm lamp, liquid crystal display and GSM900 Module, these actuators will turn on/off depending on the sensor data. The interface GSM Module SIM900 is able to connect with a cellular network. The internet of things part consist of WI-FI Module (Nodemcu-ESP8266) and Thingsboard cloud in this part we send sensor data to be stored in the cloud server so we can get it back when we need it .

Thesis layout:

The Chapters of the Research are Organized as Follows:

Chapter one gives a general overview for the system, this chapter also includes the problem statement, objectives and the methodology which was followed to solve the research problem. Chapter Two gives a summary of the background and the related literature up to the state-of-the-art regarding the problem and its possible solutions. Chapter Three shows the methodology and simulation setting up. Chapter Four involves the results which are analyzed and interpreted to evaluate the proposed algorithms and the system performance. Finally, Chapter Five the thesis is concluded in the conclusion along with suggestions for further future research.

CHAPTER TWO

BACKGROUND AND RELATED WORK

2.1 Background

An estimated average of 4,200 home structure fires per year started with the ignition of natural gas. These fires caused an average of 40 deaths per year. US Local fire departments responded to an average of 125,000 natural gas or LPG-Gas leaks per year that did not result in fires in or at home properties in 2012-2016. Since 2007, these incidents have generally been increasing. Also, gas leak can cause various health issues such as natural gas poisoning which is caused by High levels of natural gas exposure, which is characterized by fatigue, severe headaches, memory problems, loss of concentration, nausea, loss of consciousness, and suffocation. Early identification of gas leakage can prevent hazardous results, a hazard can be material, or an activity, or a procedure that can cause harm to human, or environment, or incur economical loss. After studying the related works for similar topics, the proposed system was designed. An MQ-2 and MQ-7 sensors detects the gas level and if the sensor value is greater than the threshold value then it will alarm the owner through GSM signal and turns on an alarm to alert nearby people and display the gas level on LCD to alert the users. It will turn on an exhaust ventilation fan to transfer the gas to an open space which will eventually dissolve into normal air pressure very quickly, and a relay will cut off the main power supply and the valve is turned off to prevent further leakage. To avoid any accident we find many kind of safety system.

2.2 Safety Critical System

Safety-critical systems are those systems whose failure could result in loss of life, significant property damage or damage to the environment. There are many well-known examples in application areas such as medical devices, aircraft flight control, weapons and nuclear systems. Many modern information systems are becoming safety-critical in a general sense because financial loss and even loss of life can result from their failure. Future safety-critical systems will be more common and more powerful. From a software perspective, developing safety-critical systems in the numbers required and with adequate dependability is going to require significant advances in areas such as specification, architecture, verification and the software process. The very visible problems that have arisen in the area of information system security suggest that security is a major challenge too. For examples: Circuit breaker, emergency services dispatch system, electricity exaggeration transmission and distribution, fire alarm and fire sinker[2].

Types of Safety Critical System:

2.2.1 Process Safety System or Process Shutdown System (PSD)

PSD Is defined as the automatic isolation and the activation of all part of a process, during a PSD the process remain pressurized. Basically PSD consist of field mounted sensors. Valves and trip relays, a system logic unit for processing of incoming signals, alarm and MHI units. The system is able to process all input signals and activating output in accordance with the applicable cause and effect matrix chart

Typical Action from PSD Systems:

1. Shutdown the whole process.
2. Shutdown part of the process.
3. Depressurize/Blow down parts of the process.

2.2.2 Safety Shutdown System

The safety shutdown system (SSS) shall shut down the facilities to a safe state in case of an emergency situation, thus protecting personnel, the environment and the asset. The safety shutdown system shall manage all inputs and outputs relative to emergency shutdown (ESD) functions (environment and personnel protection). This system might also be fed by signals from the main fire and gas system. it has two types:

1) Emergency Shutdown (ESD)

ESD Is designed to minimize the consequences of emergency situations, such as outbreak of fire in gas filled areas or areas which may otherwise be hazardous.

An emergency shutdown system for a process control system includes an emergency shutdown (ESD) valve and an associated valve actuator. An emergency shutdown (ESD) controller provides output signals to the ESD valve in the event of a failure in the process control system, a solenoid valve responds to the ESD controller to vent the actuator to a fail state, a digital valve controller (DVC) test strokes the ESD valve. An impedance booster device enables the dc powering of the solenoid valve and the DVC over a two wire line while still permitting digital communication over the same two wire line.

Typical Actions from an Emergency Shutdown System:

- Shutdown of part systems and equipment.
- Isolate electrical equipment.
- Prevent escalation of events.
- Depressurize / Blow down.
- Emergency ventilation control.
- Close watertight doors and fire doors.
- Centralized Project Development for Both Safety and Process needs.

2) Emergency Depressurization (EDP) System

Depressurization of any equipment must reduce its pressure to a certain amount and at a given time. This differs with the type of depressing:

High rate or low rate depressurization.

2.2.3 Detectors

They are detectors and sensors connected to the control panel, to detect the fire and gas and send a signal to the control panel. There are many types of sensors or detectors used in the alarm systems such as:

2.2.3.1 Gas Detectors

It is a device that detects the presence of gases in an area, often as part of a safety system. It works when there is a dropout of gases in the building, it detects if there are any gases in the building, and therefore sends a signal to the control panel, one of the common gas sensor detection is MQ sensors. MQ gas sensors are a family of sensors which are used to detect a wide variety of gases like alcohol, smoke, methane, LPG, hydrogen, NH₃, Benzene, Propane etc. These sensors are made up of an electrode which is coated with a sensing material, and it is heated to make it more reactive and sensitive. We use in our project MQ-2, MQ-7 gas sensors. shown Table (2-1):

Table 2-1: Gas Sensors

Sensor	Detects	Heater Voltage
MQ-2	Methane, Butane, LPG, smoke	5V
MQ-3	Alcohol, Ethanol, smoke	5V
MQ-4	Methane, CNG Gas	5V
MQ-5	Natural gas, LPG	5V
MQ-6	LPG, butane gas	5V
MQ-7	Carbon Monoxide	Alternating 5V and 1.4V
MQ-8	Hydrogen Gas	5V
MQ-9	Carbon Monoxide, flammable gasses.	Alternating 5V and 1.4V
MQ131	Ozone	6V
MQ135	Air Quality (Benzene, Alcohol, smoke)	5V
MQ136	Hydrogen Sulfide gas	5V
MQ137	Ammonia	5V

Sensor	Detects	Heater Voltage
MQ138	Benzene, Toluene, Alcohol, Acetone, Propane, Formaldehyde gas, Hydrogen	5V
MQ214	Methane, Natural gas	6V
MQ216	Natural gas, Coal gas	5V
MQ303A	Alcohol, Ethanol, smoke	0.9V
MQ306A	LPG, butane gas	0.9V
MQ307A	Carbon Monoxide	Alternating 0.2V and 0.9V.
MQ309A	Carbon Monoxide, flammable gasses	Alternating 0.2V and 0.9V
MG811	Carbon Dioxide (CO ₂)	6V
AQ-104	Air quality	

2.2.3.1.1 Gas Sensor (MQ-6)

MQ-6 gas sensor is an electronic sensor used for sensing the concentration of gases in the air such as LPG, propane, methane, hydrogen, alcohol, smoke and carbon monoxide, it is also known as Chemi-resistor. It contains a sensing material whose resistance changes when it comes in contact with the gas. This change in the value of

resistance is used for the detection of gas[3]. Shown figure (2-2):



Gas Sensor (MQ-6)

2.3.4 Alarm Unit

Used to make a sound to inform the workers in the faculty or people in the building there is emergency situation, there are many elements used in the alarm such as: Horns, Bells, Sirens and LED indicator. Shown figure (2-5)



Figure 2-5: Alarm Unit

2.4.1 Fire Sprinklers System

It is an important part in the safety system in the fire protection, it consist of a water supply and distribution system along the pipelines, it has many types like wet pipe, wet pipe antifreeze, dry pipe. Shown figure (2-6):



Figure 2-6: Fire Sprinklers System

2.4.2 WiFi Module (MODMCU ESP32)

The NodeMCU (Node MicroController Unit) is open-source software and hardware development environment built around an inexpensive System-on-a-Chip (SoC) called the ESP32. The ESP32, designed and manufactured by Espressif Systems, contains the crucial elements of a computer: CPU, RAM, networking (WiFi), and even a modern operating system and SDK. NodeMCU has 4 ground pins and 4 power pins. One is V_{in} to supply the NodeMCU directly and the other three pins are output to feed external components. It has two control pins (enable and reset), has I2C pins, GPIO pins, UART pins, PWM pins, SPI pins, ADC pins, and SDO pins [5]. Shown figure (2-12):



Figure 2-12: WiFi Module (MODMCU ESP32)

2.3.5 3.7 V Battery

The 3.7V lithium-ion battery serves as the primary power source for the entire portable gas leakage detection system, enabling it to function independently without relying on a constant mains power connection. This makes the device suitable for placement in kitchens, gas storage areas, or industrial zones where portability and continuous operation are essential. The battery provides stable power to the ESP32, sensor, buzzer, and LEDs, ensuring uninterrupted monitoring. Since 3.7V is lower than the ESP32's required voltage in some configurations, it works alongside a boost converter to step up the voltage appropriately. The rechargeable nature of lithium batteries also makes this system cost-effective and environmentally friendly, allowing repeated use through the charging module without frequent battery replacements, ideal for long-term IoT safety deployments.



2.3.6 3.7 V Battery Case

The 3.7V battery case is a protective housing designed to securely hold the lithium-ion battery in place within the device enclosure. It prevents physical damage, short circuits, and accidental disconnection of battery terminals, which is especially important in a safety-critical application like gas leak detection. The case typically includes spring-loaded contacts or wired leads that connect directly to the circuit, ensuring stable and consistent power delivery to the ESP32 and other

components. Using a proper battery case also simplifies maintenance, as the battery can be easily removed for charging or replacement without disturbing the rest of the circuit. Additionally, it enhances the overall safety and durability of the project by insulating the battery from dust, moisture, and accidental impacts, especially when installed inside a compact 3D-printed enclosure.



2.3.7 Step-Up Boost Converter

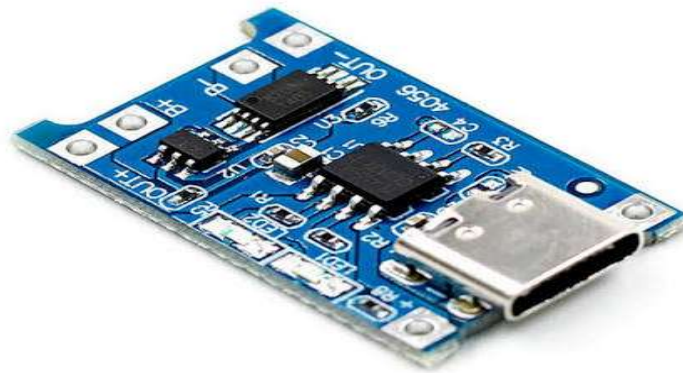
The step-up boost converter plays a critical role in regulating power for the system, as the 3.7V output from the lithium battery is insufficient to reliably power the ESP32 and peripheral components, which typically require 5V. This module boosts the lower battery voltage up to a stable 5V output, ensuring consistent performance of the ESP32, sensor, and buzzer even as the battery voltage gradually drops during discharge. Without this converter, the system could behave erratically or shut down prematurely as battery voltage decreases. The boost converter thus extends the usable battery life while maintaining reliable operation, making it an essential bridge

between the raw battery power and the voltage requirements of the electronic components used throughout this compact and portable gas leakage detection device.



2.3.8 Charging Module

The charging module, often a TP4056 or similar circuit, is responsible for safely recharging the 3.7V lithium-ion battery once it has been depleted. It regulates the charging current and voltage, protecting the battery from overcharging, overheating, or short circuits, which are critical safety considerations when working with lithium batteries. This module typically includes a micro-USB or USB-C input port, allowing convenient charging from any standard power adapter or power bank. By incorporating this module, the gas leakage detector becomes a reusable, sustainable device rather than requiring disposable batteries. It also often includes status LEDs indicating charging progress and completion. This feature significantly enhances the practicality of the project, making it suitable for long-term deployment in homes or industrial settings without frequent maintenance.



2.3.9 Jumper Wires

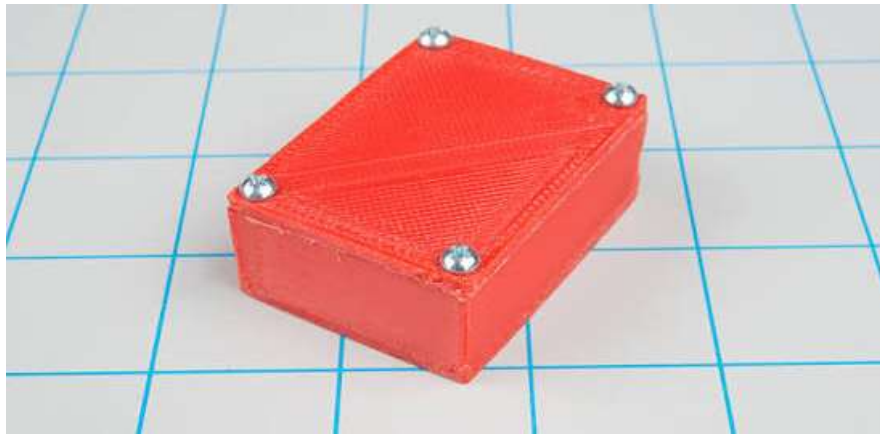
Jumper wires are essential connecting elements used throughout the circuit to establish electrical connections between the ESP32, MQ-6 sensor, buzzer, LEDs, resistors, switch, and power modules.

Available in male-to-male, male-to-female, and female-to-female variants, they allow flexible and tool-free wiring on breadboards or directly between component pins. In this project, jumper wires link the sensor's analog output to the ESP32's ADC pin, connect the buzzer and LEDs to designated GPIO pins through resistors, and route power from the battery and boost converter to all active components. Their flexibility simplifies prototyping and testing, allowing quick modifications or debugging before finalizing the circuit inside the 3D-printed enclosure. Reliable, well-insulated jumper wires ensure stable signal transmission and minimize the risk of loose connections or circuit failure.



2.3.10 3D Printed Box

The 3D printed enclosure serves as the protective housing for the entire gas leakage detection system, holding the ESP32, sensor, buzzer, LEDs, battery, and other components securely in one compact unit. Custom-designed and 3D printed, this box can be tailored precisely to the dimensions of each component, ensuring a neat, organized, and professional-looking final product. It typically includes ventilation slots to allow gas to reach the MQ-6 sensor freely, along with cutouts for the buzzer sound, LED visibility, switch access, and charging port. This enclosure not only improves the aesthetic appeal of the project but also protects sensitive electronics from dust, accidental impacts, and moisture. Being lightweight and customizable, 3D printing allows rapid prototyping of different enclosure designs, making the final device durable, portable, and suitable for real-world installation.



2.3.11 220 Ohm Resistor(x2)

Two 220-ohm resistors are used in this circuit primarily to limit current flowing through the red and green LEDs, protecting them from damage due to excessive current draw from the ESP32's GPIO pins. Without these resistors, the LEDs could burn out quickly or draw more current than the microcontroller pins can safely supply. Each resistor is placed in series with its respective LED, ensuring appropriate brightness while maintaining component longevity. The 220-ohm value is a standard choice that balances adequate current

limiting with sufficient LED brightness for clear visual indication. These resistors are simple yet crucial passive components that safeguard the visual alert system, ensuring the red LED (danger indicator) and green LED (safe status indicator) function reliably throughout the operational lifespan of the gas leakage detection device.



2.3.12 Switch

The switch in this project serves as the primary power control mechanism, allowing users to manually turn the entire gas leakage detection system on or off. Positioned between the battery power source and the rest of the circuit, it provides a simple and convenient way to conserve battery life when the device is not actively needed, or to reset the system if required. Including a physical switch enhances user control and safety, as it prevents unnecessary battery drain during storage or transport. It also allows for quick troubleshooting, letting users power-cycle the device without needing to disconnect wiring manually. This small but functional component adds practicality and user-friendliness to the overall design, making the device easier to manage in everyday household or industrial safety applications.



2.3.13 Red LED and Green LED

The red and green LEDs serve as visual status indicators for the gas leakage detection system, providing an immediate, easy-to-understand representation of the environment's safety condition. The green LED remains illuminated during normal operation, signaling that gas levels are within safe limits and the area is secure. When the MQ-6 sensor detects gas concentration exceeding the safety threshold, the ESP32 switches off the green LED and activates the red LED alongside the buzzer, clearly signaling danger. This dual-LED system offers an intuitive, at-a-glance safety status, useful even for those who may not hear the buzzer or are located at a distance. Together with the audible alarm, these LEDs strengthen the overall alert mechanism, making the gas leakage detector both effective and accessible for everyday use.



2.4.3 Buzzer

A buzzer or beeper is an audio signaling device; it is generally used as sound alarms. It generates sound in a frequency range of 1 to 7 kHz as an audio indication. In this frequency range, the hearing threshold is at maximum. Shown figure (2-18):



Figure 2-18: Buzzer

2.5 Internet of Things (IoT)

The Internet of things refers to a type of network to connect anything with the Internet based on stipulated protocols through information sensing equipment to conduct information exchange and communications in order to achieve smart recognition, positioning, tracing, monitoring, and administration. Internet of things (IoT) is a global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies. The IoT improve the safety of the factory by monitoring Possible Risks by analyzing data and using it to develop occupational health and safety strategies for employees and increase understanding of working environments with the help of connected sensors[9].

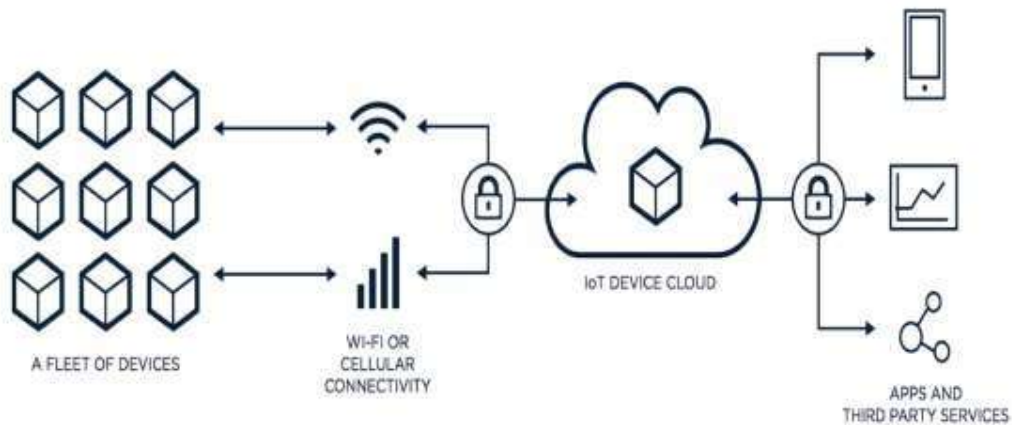


Figure 2-21: Internet of Things (IoT) System

2.6 The Cloud

"The cloud" refers to servers that are accessed over the Internet, and the software and databases that run on those servers. Cloud servers are located in data centers all over the world. By using cloud computing, users and companies do not have to manage physical servers themselves or run software applications on their own machines[10]. Shown figure (2-22):

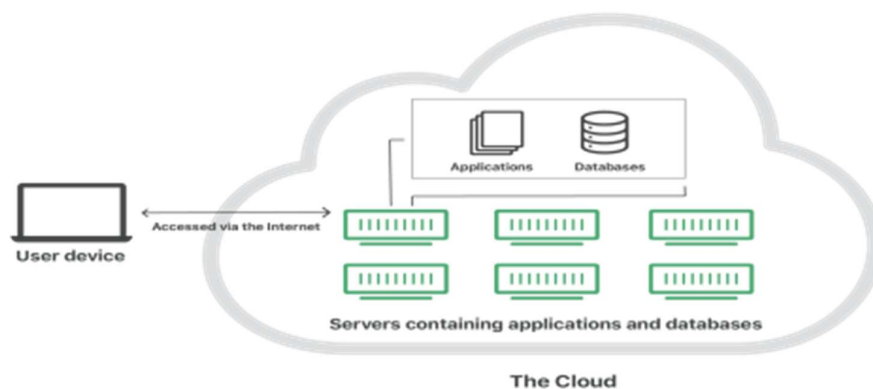


Figure 2-22: The Cloud

The cloud enables users to access the same files and applications from almost any device, because the computing and storage takes place on servers in a data center, instead of locally on the user device.

The Main Service Models of Cloud Computing:

Software-as-a-Service (SaaS): Instead of users installing an application on their device, SaaS applications are hosted on cloud servers, and users access them over the Internet. Examples of SaaS applications include Sales-force, Mail-chimp, and Slack.

Platform-as-a-Service (PaaS): In this model, companies don't pay for hosted applications; instead they pay for the things they need to build their own applications. PaaS vendors offer everything necessary for building an application, including development tools, infrastructure, and operating systems, over the Internet. PaaS examples include Heroku and Microsoft Azure.

Infrastructure-as-a-Service (IaaS): In this model, a company rents the servers and storage they need from a cloud provider. They then use that cloud infrastructure to build their applications. IaaS providers include Digital-ocean, Google Compute Engine, and Open Stack.

In Recent Years a Fourth Model has Emerged:

Function-as-a-Service (FaaS): FaaS, also known as server less computing , breaks cloud applications down into even smaller components that only run when they are needed. Shown figure (2-23):

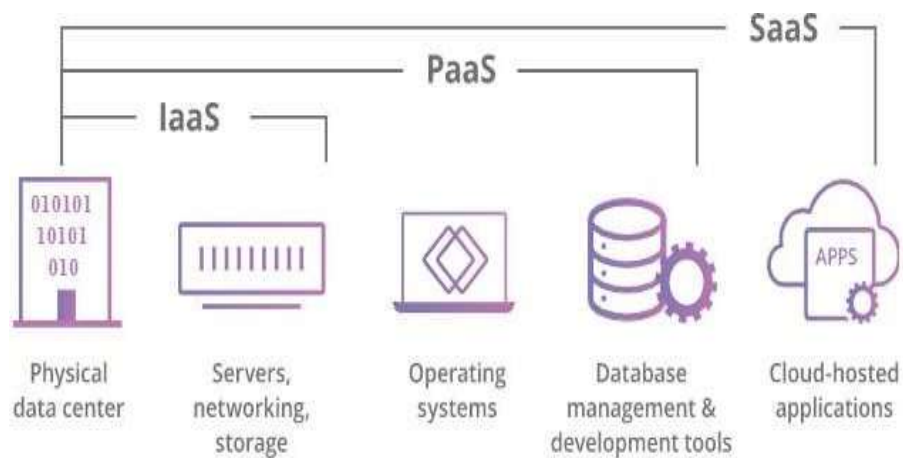


Figure 2-23: Models of Cloud

2.6.1 Things-board Cloud

Things-Board is an open-source IoT platform that enables rapid development, management and scaling of IoT projects. It enables device connectivity via industry standard IoT protocols - MQTT, CoAP and HTTP and supports both cloud and on-premises deployments. Things-Board combines scalability, fault-tolerance and performance so we will never lose our data[11]. Shown figure (2-24):

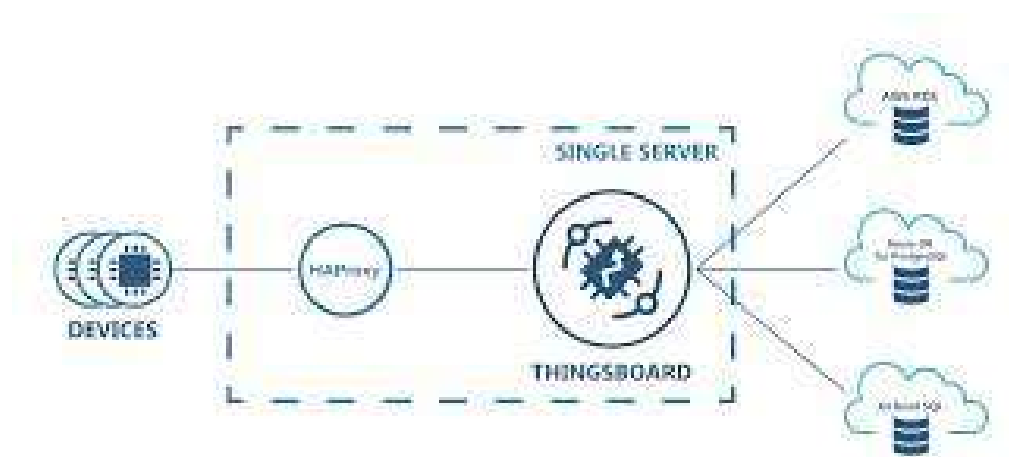


Figure 2-24: Things-board Cloud

2.7 Related Work

Many similar projects already exist and they have their own characteristics and functionality. They all have so many advantages as well as disadvantages. We can see various descriptions of them below here.[12]Manish Verma used Raspberry pi 3 model attached with embedded system with required input and output gas level with the help of gas sensors. This resulted in a more efficient in operation because it is connected to a common web page specially built to notify or email the responsible authority automatically so reduces the stress of constant monitoring. The choice of using a real time gas leakage monitoring and sensing the output levels of gas has been clearly observed by the help of this system.

This project[12] implements a model which sends an email to the user in case there is a leakage. The model detects the leakage of Liquid Petroleum, Gas, & Benzene, this cost-effective project uses MQ-6 and MQ-135 sensors for detecting the aforementioned gases using Arduino - UNO, WIFI Module ESP8266 and Thing-speak cloud. The real-time information of the above-mentioned gases are uploaded in the cloud and displayed in the form of a graph to the user. The prototype of this model generates an email to the concerned person using IFTTT web service. An LED is also used as a visual alarm at the site of leakage.

The authors in[13] developed a system to monitor and control the gas leakage concentration. MQ-6 gas sensor is used for sensing the level of gas concentration in a closed volume. To monitor the consequences of environmental changes an IoT platform hosted by “Things-speak” platform has been introduced. Both robust and cloud-forwarded controls have been applied to prevent uncontrolled leakage of those gases and

auto-ignition. This type of system can be directly applied to the engine chamber/ fuel chamber of the modern marine vessels using dual fuel power cycle with LPG/LNG as secondary fuel-flamer. The results from the experiments clearly indicate satisfactory actuation speed and accuracy. The trials performed by the authors showed about 99% efficiency of signal transmission and actuation.

The authors of this paper[14] presented a project that deals with monitoring LPG leakage along with administrative alert by giving buzzer sound, switching on specified relay(s) and sending an alert message to administrator(s) to decide about the precaution measures. For this purpose, gas leakage concentration is sensed by gas sensor (MQ-6) which sends the data (analog) to the controller (NodeMCU) where the analog value is subjected to a sequential conversion to predict the probable intensity of gas inside the control volume and on crossing a reference threshold value (set by administrator/operator) the controller it switches „ON“ the relay(s) and buzzer(s) alerting to the administrator/operator of the plant (Industry/Home) including an extra control-option through manual control utilizing the IoT server to prevent accidents from the leakage.

This gas detection and alert system[15] will not only alert us of the leakage but will also mechanically turn off the knob of the gas cylinder to seize any leakage of gas. Moreover, this system informs the concerned person by emailing and dropping a message on their mobile. This gives a larger degree of safety to any gas setup in any circumstances. The project is determined by using a lighter to collect leaked gas around the gas sensor after sensing procedure if sensor value is greater than the threshold value then microcontroller will perform its programmed tasks:

- 1.Immediately turn off the regulator knob to stop further leakage
- 2.Within 2-4sec the relay will cut off the main power supply.
- 3.Buzzer starts beeping and a message is displayed on LCD to alert the users and nearby people.
- 4.Wi-Fi module will send SMS/e-mail using the cloud to the users.
- 5.The exhaust fan will fan out all enclosed gas from the environment. When reset button of MCU pressed the system will get refreshed

As a control framework, the authors of this paper[16] use Raspberry pi, MQ gas sensors, and an alert circuit. With the guide of Things-speak cloud benefits, the yield of the data gathered by the sensors is stored into the cloud utilizing the Distributed computing (IoT). The technique for distinguishing conceivably unsafe gas spills by sensors and directing the bot with the guide of catches is UGV Gas spill discovery. Ordinarily, these gadgets utilize a hear-able bell to caution people that a hurtful gas has been recognized. Utilizing the MQ sensors interconnected to the Raspberry pi, this recognizable proof can be cultivated. A voltage is delivered in it at whatever point the framework recognizes gas and is provided as a contribution to the raspberry pi. At the point when the gas has been discovered, the ringer sounds. This information is saved in the cloud utilizing IoT. By means of prearranging a python code and downloading the vital sensor libraries, the whole activity of the gadget can be refined.

In this paper[17], we built an IoT (Internet of Things) based framework that helps to prevent accident-related to LPG leakage. The device is an intelligent piece of home computerization framework, as at the point whenever gas spillage is detected it alarms the customer through GSM signal phone call and SMS. All the while, it sounds the buzzers to alert nearby people and it turns on the exhaust fan to transfer the gas in an open space which will eventually dissolve into normal air pressure very quickly. The device is capable of identifying the gas and fire separately so that it can send accurate information directly to the user via GSM module as quickly as possible. The algorithm is designed to detect and produce results as quickly as accurately possible to prevent any accident/blast from happening.

When a gas leakage occurs in an underwater structure[18], the high-pressure gas jetted from the leak point will form bubbles in the nearby waters, and the leak point will produce radiation noise with a sound source level much higher than the environmental background and certain characteristics. Active sonar detection technology uses image sonar to detect whether there is a bubble group in the target area to determine whether there is a leakage and the passive sonar detection technology uses whether the radiation noise intensity is stronger than the background noise to determine whether there is a leakage. Compared with other detection methods, the sonar detection technology is an external inspection method. It does not need to install the measuring equipment on the measured object in advance, and only needs to deploy the sonar array in the detection area, which has the advantages of easy expansion and easy installation. In addition, compared to active sonar, passive sonar does not generate acoustic signals to interfere with other

acoustic equipment used in the oil production, and passive sonar systems have lower power consumption and longer lifespan.

This paper[19] presents the design and development of a wireless gas leakage monitoring system by using Arduino and Zigbee. In this project, the monitoring system is developed by using LabVIEW GUI. It is used to display the level of gas concentration in a place through another remote PC, and via internet server. Hence, it provides benefit to monitor the condition of a room in a safe distance. Traditionally, the gas pipeline leakage monitoring system is realized by communication cable system, therefore the cost of installation and maintenance are very expensive and difficult as mentioned by jDing . In order to overcome these restrictions, wireless sensor network is chosen as the best choice in the situation above. Some papers proposed different types of wireless sensor network such as radio frequency (RF) transceiver router and coordinator, general packet radio service (GPRS) network and Zigbee. Nowadays, Zigbee is widely used in the gas leakage monitoring application field for the real-time monitoring of the potential risk areas. For the autonomous control system is as a preventive way to stop the situation becoming worst by shutting down the process automatically. A. Shrivastava has proposed the system by using stepper motor to turn off the main power and the gas supply. Whereas in this model, the relay switch is used to turn off the main power, and the electronic gas valve is used to turn off the gas supply. At the end, when the gas leakage is successfully stopped then the whole system will return to initial stage with the help of reset button.

CHAPTER THREE

METHODOLOGY

In this chapter we will discuss the problem solution and the system components and discuss about each component and its use in the system.

3.1 Overview

The design of the system consist of three parts, first part is to sense the gas concentration, fire and temperature measurement. The second part is to monitor the factory environment using an application, the third part is to control the system through Arduino mega and relays for the run-on and run-off of the panel and water pump and automatic valve.

The proposed system working method start by receiving signals from the sensors and in the case of a gas leak or fire accident the Arduino determines the situation based on the sensors data, if there was gas leak the system will turn off the gas valve using solenoid valve and turn on the ventilation fans to reduce the gas concentration by using a mechanical relay and send a SMS to the managers informing them , when contacting the gas and an alarm will start to notify people in the factory, the alarming part consist of a buzzer and LCD that inform the worker there is gas leak. And in the case of fire detection the system will turn off the gas valve and turn on the fire-fighting system and ventilation fans to reduce the smoke and call the managers to inform them there is a fire and the factory power will shut-down and the alarm system will be activated. All the sensors data that received by Arduino will be saved on

the cloud or database and used to monitor the system through the application that act like human machine interface (HMI), we can save this data in the cloud by connecting Arduino to the internet by using ESP8266. And we use SIM900 to send SMS or make a call to inform the manager the emergency situation. The system block diagram is shown in figure (3-1):

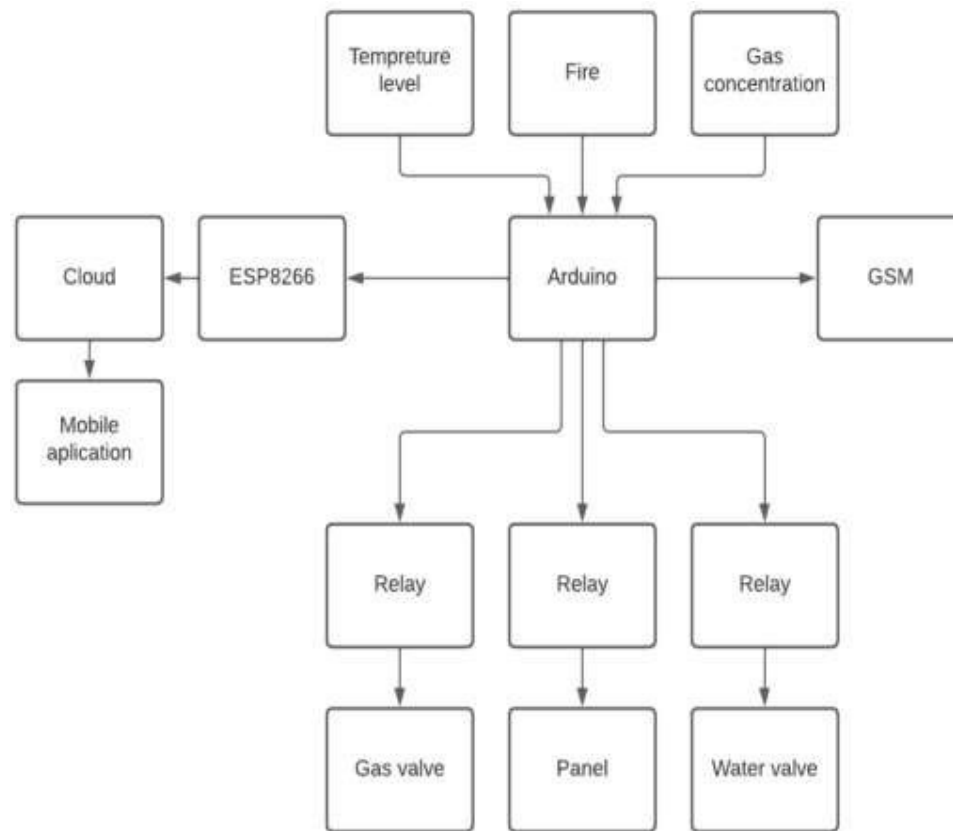


Figure 3-3: System Block Diagram

3.2 Simulation

The following figure (3-2) illustrates the system circuit for IoT gas and fire detection which has been designed using the Proteus program , We Considering that the factory is divided into two part each part has LPG gas sensor (MQ-2), CO gas sensor (MQ-7), flame sensor and

temperature sensor ,so when the system detect any gas or fire in any part in the factory it will decide the right action using the actuators.

3.3 System Interfacing

3.3.1 Interfacing Gas Sensor (MQ-6) with Arduino

The gas sensors are connected to the Arduino Mega which is supplied by 5 DC voltage from the regulator by wiring VCC and GND pins of gas sensors to VCC and GND pins of Arduino and then the signal pin of gas sensors are connected to the analog pin in the Arduino (53) for the first MQ-6 and pin (52) for the second MQ-6 and pin (51) for the first MQ-6 and pin (50) for the second MQ-6.

3.3.2 Interfacing Flame and Temperature Sensors with Arduino

The Arduino Mega is supplied by 5 DC voltage from the regulator, then the gas sensors are connected to the Arduino by wiring VCC and GND pins of gas sensors to VCC and GND pins of Arduino and then the signal pin of gas sensors are connected to the analog pin in the Arduino (13) for flame one and pin (11) for flame two and pin (12) for temperature one and pin (10) for temperature two.

3.3.3 Interfacing the First Relay Circuit with Arduino

This circuit controls the gas valve, the circuit construction from relay RL1, it has five pins. Two are connected to the Arduino pin (6) and ground (control pins), the third pin is connected to the gas valve negative point and the fourth pin is connected to the power supply which feed the gas valve positive point by 5 DC voltage and the fifth pin configuring the mode relay if it is close or open.

3.3.4 Interfacing the Second Relay Circuit with Arduino

The second relay is used to control the LED which is used in the simulation to represent the electricity panel, the circuit construction from relay RL2, it has five pins. Two are connected to the Arduino pin (8) and ground (control pins), the third pin is connected to the LED negative point and the fourth pin is connected to the power supply which feed the LED positive point by 5 DC voltage and the fifth pin configuring the mode relay if It is close or open.

3.3.5 Interfacing the Third Relay Circuit with Arduino

This circuit controls the motor which represent the fire-fighting system, which is controlled by the relay RL3, it has five pins. Two are connected to the Arduino pin (7) and ground (control pins), the third pin is connected to the motor negative point and the fourth pin is connected to the power supply which feed the motor positive point by 5 DC voltage and the fifth pin configuring the mode relay if it close or open.

3.3.6 Interfacing the Liquid Crystal Display (LCD) with Arduino

The VSS pin of the LCD is connected to the ground, while the VDD pin is fed by 5 DC voltage, the VEE pin is connected to the monitor with variable screen-lighting control resistance, the RW pin is

connected to the ground, the RS pin is connected to the seventh digital pin of Arduino, the E (enable) is connected to sixth digital pin of Arduino, the data legs (D4-D7) are connected with pins (25 -22) of Arduino.

3.3.7 Interfacing the Ventilation Fan with Arduino

The ventilation fan positive input is connected to the Arduino pin (2) and the negative pin is connected to the ground, it will be active when it receives a signal from Arduino to reduce the gas concentration or smoke.

3.3.8 Interfacing the Buzzer with Arduino

It is connected to the Arduino pin (5) and the other side is connected to the ground, it will be active when it receives a signal from Arduino to alarm the people in the factory.

3.3.9 Interfacing the Light Emitting Diode (LED) (Red, Green) with Arduino

It is connected with a 450 ohm resistance and the resistance is connected with Arduino pin (4) for the red LED and Arduino pin (3) for the green LED, the other side of LED is connected with the ground when the red LED receive a signal from the Arduino that mean the system is in the emergency mode and when the green LED is active that mean the system in the safe mode.

3.3.10 Interfacing the Wireless Module (GSM900) with Arduino

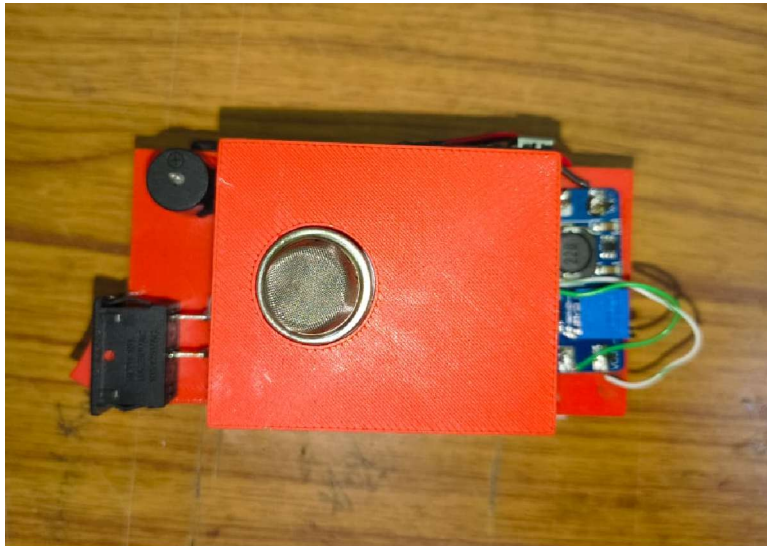
The GSM900 TX pin is connected to the Arduino through RX pin and its RX pin is connected to the TX pin in Arduino, a virtual terminal

RX and TX are connected to the TX and RX pins of the GSM900 respectively to display the messages from the GSM.

3.4 System Operation

First of all, all the hardware unit of the system was tested and it was ensured that they were in a good working condition. Then all units, gas valve, electrical panel, fire-fighting system, buzzer, ventilation fan, gas sensors, flame sensors, temperature sensors, LCD and LEDs, were connected to each other on the board with the Arduino Mega using wires, the sensors were connected to 5v from the Arduino, the actuators were connected to a 12v external power supply and the AC contractor is connected to a 24v external power supply, the system response is divided into two parts: normal mode “no gas or fire detected” and emergency mode “gas or fire detected”. In the normal situation the green LED light up, and the LCD screen displays the phrase „safe“ the gas valve relay and electricity panel relay are normal close, the fire-fighting system relay is normal open and the ventilation fan is close. In the emergency mode if one of the gas sensors is sensing (LPG or CO) gas the sensor sends the signal by analog signal pin to the Arduino, then the Arduino receives and read the signal and compare it with the threshold value, if it is higher than threshold value the Arduino send a high digital signal that will change the gas valve relay from normal close to open to turn off the valve, and send a high digital signal to the ventilation fan, buzzer and red LED to be active. and if one of the flame sensors is active then the sensor sends the signal by digital signal pin to the Arduino, then the Arduino receive and read the signal if it is equal to zero the Arduino send a high digital signal that will change the fire-fighting system relay from normal open to close, and change the gas valve relay from normal close to open to turn off the valve, and send a high digital signal to the

ventilation fan, buzzer and red LED to be active then send a high digital signal that will change the solid state relay from normal close to open to turn off the AC contactor to shut-down the factory power. Second part is the software, the code used to program the system, after importing the required libraries and declaring the variables, the values from the sensors are read using specific functions like Digital Read and AnalogRead, and then stored in the variables, next the sensor value is tested using an IF statement and if value is above the predefined value then the Arduino will send controlling signal to the actuators accordingly, if a gas leakage is detected a SMS is sent using send SMS function, and in the case of fire a call is made using call function function.



IoT-Based Smart Gas Leakage Detection and Emergency Alert System

3.1 System Block Diagram

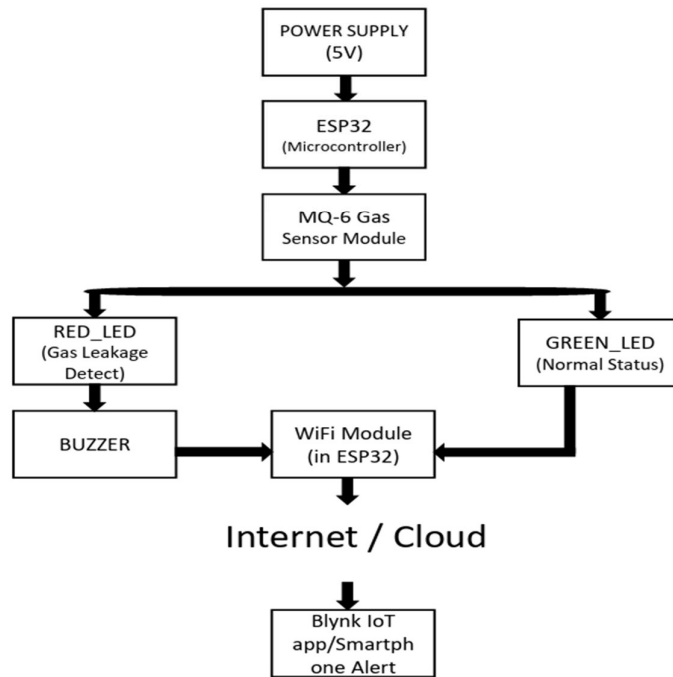
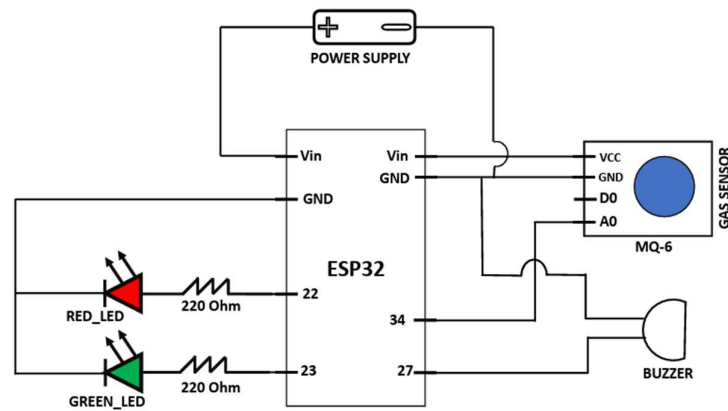


Figure 3-3: System Block Diagram

CHAPTER FOUR

SYSTEM RESULT

4.1 System Operating Condition

4.1.1 Fire is Detected

If both of the flame sensor or one of them detected a fire ,the sensor will send signal to the Arduino so the Arduino sends a signal to: the first relay to operate the fire-fighting system, to the second relay to close the gas valve, to the fan to reduce the smoke, to the alarming part (buzzer and LED),to the LCD to display the case, to the GSM to make a warning call to the manager, and to the electrical panel to shut-down the factory power.

4.1.2 Gas is Detected

We have 4 gas sensors two of them are MQ-2 and the other two are MQ7- when these sensors detect gas then it will send signal to the Arduino and therefore the Arduino sends a signal to: the second relay to close the gas valve, to the fire-fighting system to reduce the concentration of the gas, to the alarming part (buzzer and LED), to the LCD to display the case, and to the GSM to send a warning message to the manager.

4.1.3 Fire and Gas is Detected

Any one of the flame sensor with any one of the gas sensor are active then the sensor will send signal to the Arduino then the Arduino sends a signal to: the first relay to operate the fire-fighting system, to the second relay to close the gas valve, to the ventilation fan to reduce the concentration of the gas, to the alarming part (buzzer and LED), to the

LCD to display the case, and to the GSM to send a warning message and call the manager.

4.1.4 No Detection

The system is in the normal mode the sensor voltage is less than the threshold so the electricity panel and gas valve are active and the fan, water pump and alarming unit are off.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

In this thesis, the aim is to design a gas and fire alarm and control system with a Low cost with effective usage and make it more users friendly and easy to operate. So gas, temperature and flame sensors and Arduino are used to save lives and reduce percentage of any accident. The program embedded in the Arduino Mega works according to the need A step by step approach in designing Arduino based system for gas and temperature measurement has been followed. According to the study and analysis of various parts of the system a design has been carried out the results obtained from the measurement has shown that the system performs well under all conditions and the attempt has been done

5.2 Recommendation

From the outcomes of the research, the following recommendations can be made:

1. Predicting the occurrence of a gas leak or fire based on AI which learning form data we storage in our cloud
2. In case the system breakdown , by using NODEMCU WI-FI module and GSM we can send a message to the manager that the system breakdown and the sensors have no read.

APPENDIX

/*

Gas Leakage Protection System (Skeleton)

NOTE:

This file is a ready starting point that fixes the blocking Wi-Fi logic.

Fill in your WiFi and HiveMQ credentials where indicated.

Hardware:

MQ2 AO -> GPIO34

RED LED -> GPIO25

GREEN LED -> GPIO26

BUZZER -> GPIO27

*/

#include <WiFi.h>

#include <WiFiClientSecure.h>

#include <PubSubClient.h>

const char* ssid = "moto edge 60 fusion";

const char* wifi_password = "WahShampiWah";

```

const char* mqtt_server =

"0b7a0207350e48139ae6644afb704499.s1.eu.hivemq.cloud";

const int mqtt_port = 8883;

const char* mqtt_username = "gasproject";

const char* mqtt_password = "PritamGasproject@2003";


#define MQ_PIN 34

#define RED_LED 22

#define GREEN_LED 23

#define BUZZER 27


const char* topic_value="pritam/gas/value";

const char* topic_status="pritam/gas/status";

const char* topic_alert="pritam/gas/alert";


WiFiClientSecure secureClient;

PubSubClient client(secureClient);


int gasThreshold=1500;

bool alertSent=false;

unsigned long lastSample=0;


void wifiTask(){

    static unsigned long lastTry=0;

    if(WiFi.status()==WL_CONNECTED) return;

    if(millis()-lastTry<5000) return;

```



```

    lastTry=millis();

    WiFi.begin(ssid,wifi_password);

}

void mqttTask(){

    static unsigned long lastTry=0;

    if(WiFi.status()!=WL_CONNECTED) return;

    if(client.connected()) return;

    if(millis()-lastTry<5000) return;

    lastTry=millis();

    String id="ESP32Gas-

"+String((uint32_t)ESP.getEfuseMac(),HEX);

    client.connect(id.c_str(),mqtt_username,mqtt_password);

}

void publishValue(const char* t,const char* p){

    if(client.connected()) client.publish(t,p);

}

void setup(){

    Serial.begin(115200);

    pinMode(RED_LED,OUTPUT);

    pinMode(GREEN_LED,OUTPUT);

    pinMode(BUZZER,OUTPUT);

    pinMode(MQ_PIN,INPUT);

```

```

secureClient.setInsecure();

client.setServer(mqtt_server,mqtt_port);


// Hardware self-test

digitalWrite(RED_LED,HIGH); delay(1000);

digitalWrite(RED_LED,LOW);

digitalWrite(GREEN_LED,HIGH); delay(1000);

digitalWrite(GREEN_LED,LOW);

digitalWrite(BUZZER,HIGH); delay(500);

digitalWrite(BUZZER,LOW);

}


void loop(){

  wifiTask();

  mqttTask();

  if(client.connected()) client.loop();


  if(millis()-lastSample>=1000){

    lastSample=millis();


    int gas=analogRead(MQ_PIN);

    Serial.println(gas);


    char buf[12];

    sprintf(buf,"%d",gas);

    publishValue(topic_value,buf);

```

```

if(gas>=gasThreshold){

    digitalWrite(RED_LED,HIGH);

    digitalWrite(GREEN_LED,LOW);

    digitalWrite(BUZZER,HIGH);

    publishValue(topic_status,"LEAK");

    if(!alertSent){

        publishValue(topic_alert,"WARNING! GAS LEAKAGE
DETECTED");

        alertSent=true;

    }

}else{

    digitalWrite(RED_LED,LOW);

    digitalWrite(GREEN_LED,HIGH);

    digitalWrite(BUZZER,LOW);

    publishValue(topic_status,"SAFE");

    alertSent=false;

}

}

}

```

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